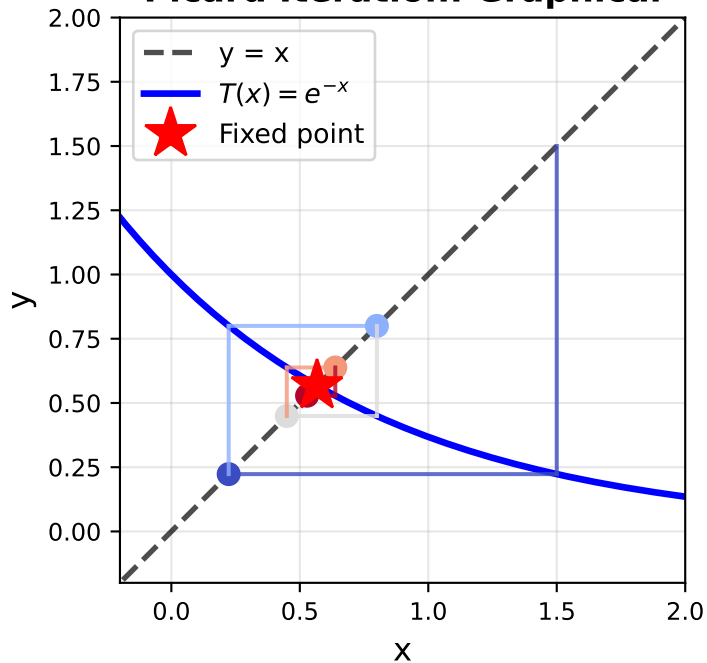


Picard Iteration: Graphical



PICARD ITERATION ALGORITHM

Input: Complete metric space (X, d) ,
contraction mapping $T: X \rightarrow X$ with constant $\alpha < 1$,
starting point $x_0 \in X$, tolerance $\varepsilon > 0$

Output: Approximate fixed point x^*

Algorithm:

1. Set $n = 0$
2. While $|x_{n+1} - x_n| > \varepsilon$ do:
 - a) Compute $x_{n+1} = T(x_n)$
 - b) If $|x_{n+1} - x_n| \leq \varepsilon \cdot (1 - \alpha) / \alpha$ then
return x_{n+1} (converged)
 - c) Set $n \leftarrow n + 1$
3. If n exceeds max iterations:
warn "Convergence may not be achieved"

Properties:

- Guaranteed convergence
- Exponential error decay
- Error estimate: $|x_n - x^*| \leq \alpha^n |x_0 - x^*|$